Local Smoothing Does Not Establish Extrapolation: A Reproducible Toy Study

Super Writer worked example

September 2026

Abstract

Better error on a familiar input range does not establish robustness outside that range. We illustrate this distinction with an executable, synthetic nearest-neighbor regression study. One-neighbor and five-neighbor predictors are fitted to 40 noisy observations from a known one-dimensional function, using five fixed training seeds. On an interior evaluation grid, mean squared error is 0.1083 for one neighbor and 0.0851 for five neighbors; on an extrapolation grid, the corresponding errors are 2.6891 and 3.7012. These measurements support a local comparison under the stated protocol, not a claim of universal superiority or statistical significance. This is an educational writing example, not a novel method or a submitted paper.

1 Introduction

A results paragraph can be numerically correct and still promise more than its experiment establishes. Consider a predictor that reduces error on a fixed evaluation set. That observation answers a question about the tested inputs, training procedure, and metric. It does not automatically answer whether the predictor improves on a different domain, under a different noise process, or with a different training budget. A useful manuscript makes that distinction visible before interpreting the result.

We use nearest-neighbor regression to make this writing problem inspectable. The estimator is standard, so the reader need not disentangle a new method from a new claim. Our question is deliberately narrow: under one specified synthetic protocol, does averaging five nearby training responses improve mean squared error both inside and beyond the training input range?

The example has two deliverables. First, it provides a short experiment that reconstructs every table entry from fixed seeds. Second, it shows how a paper can report an improvement and an adverse result without suppressing either. Neither deliverable is presented as a research novelty or evidence that this writing workflow improves conference acceptance rates.

2 Background and Scope

Nearest-neighbor regression predicts a response by averaging training responses near a query. It is a conventional local estimator, discussed alongside the bias-variance tradeoff in statistical learning texts [Hastie et al., 2009]. We use this reference for the method's background, not as support for the numerical results of this example. Those results come only from the accompanying executable.

Increasing the neighborhood size changes the observations being averaged. It can reduce sensitivity to individual noisy responses while also smoothing local variation. The balance depends on the signal, noise, sample size, and evaluation distribution. We therefore do not begin with the assumption that five neighbors must be preferable. Both neighborhood sizes and both evaluation domains are fixed in the script; there is no automated hyperparameter search.

3 Experimental Protocol

The underlying response is

$$f(x) = 2x + \sin(4\pi x).$$

For each seed in $\{11, 22, 33, 44, 55\}$, we draw 40 training inputs uniformly from $[0, 1)$ and add independent uniform response noise in $[-0.5, 0.5]$. The random generator is Python's seeded `random.Random`. Each training pair consumes one input draw followed by one noise draw. The script specifies this order so reproductions do not depend on an unstated sampling convention.

For $k \in \{1, 5\}$, prediction is the arithmetic mean of the responses at the k closest training inputs, using absolute input distance. Ties are resolved by training-row order. Both predictors see exactly the same training set within a seed. We do not learn a distance metric, rescale the input, or tune k using the evaluation results.

The in-domain grid is $x_i = (i + 0.5)/101$ for $i = 0, \ldots, 100$. The extrapolation grid is obtained by adding one to every grid point. Evaluation targets are the noiseless function values $f(x_i)$, rather than new noisy observations. Thus the reported error measures recovery of the chosen function, not prediction of future noisy labels.

For each training seed, method, and domain, we compute mean squared error over the 101 grid points. We then report the arithmetic mean and sample standard deviation across the five training seeds. The grid points are not treated as 101 independent training replications. The standard deviation describes variation across these five fits; it is not a confidence interval.

4 Results

Evaluation domain	Neighbors	Mean MSE	Sample SD
In-domain	1	0.1083	0.0056
In-domain	5	0.0851	0.0128
Extrapolation	1	2.6891	1.4620
Extrapolation	5	3.7012	1.3386

Inside the training input range, the five-neighbor predictor has lower mean MSE than the one-neighbor predictor under this protocol. The absolute reduction is approximately 0.0232. This supports the statement that local averaging improves the observed mean error in this particular comparison. We do not describe that reduction as statistically significant: five fixed training replications and a descriptive table are not an inferential test.

Beyond the training range, the ordering reverses. Five neighbors have higher mean MSE, by approximately 1.0121. Omitting this domain would leave the impression that the in-domain improvement establishes a broader property than was tested. Reporting both domains instead makes the claim's boundary explicit: the observed advantage does not carry over to this extrapolation setting.

The per-seed CSV is part of the example, alongside the aggregation JSON and the exact executable. Readers can inspect individual fits rather than rely only on rounded table entries. Reproduction checks compare unrounded values with a small floating-point tolerance and separately check the displayed rounding; they do not infer scientific validity from file existence.

5 Interpretation

For any query to the right of all training inputs, the ordering of input distances is fixed. For a fixed training set, each nearest-neighbor predictor therefore returns a constant throughout our extrapolation interval: the response at the rightmost training input for $k = 1$, or the mean response of the five rightmost inputs for $k = 5$. Neither estimator extrapolates the chosen signal's linear trend or oscillations. This follows from the estimator definition, independently of the observed ranking in the results table.

That observation explains why extrapolation needs its own evaluation; it does not prove that one neighbor will always outperform five there. The ranking also depends on noisy boundary responses, the sampled training inputs, and the target function. A different function or noise realization could change it. The strongest defensible conclusion remains a comparison of the reported errors under the stated conditions.

6 Limitations

The study uses one function, one sample size, one noise distribution, two neighborhood sizes, and five fixed seeds. It contains no real-world data, external benchmark, validation-selected hyperparameters, uncertainty-calibrated prediction intervals, or evidence about other regressors. A smooth parametric baseline could answer a different and useful question, but none was evaluated here. It must not appear in the paper as a completed experiment.

The protocol was designed for an inspectable educational example, not preregistered research. The example's author reviewed the resulting table before writing this manuscript. We make no claim that the setup or reported conditions were selected independently of the pedagogical objective.

The manuscript was authored with AI assistance for this repository and checked against its accompanying code and materials. Its original prose and synthetic experiment are public teaching artifacts, not an autonomous-writing benchmark, a blind human evaluation, or a paper accepted by any venue.

7 Conclusion

The five-neighbor predictor improves mean error on the tested in-domain grid but worsens it on the tested extrapolation grid. Keeping both observations in the manuscript turns a broad claim of robustness into a specific, reproducible comparison. The example illustrates an evidence boundary; it does not introduce a new learning method or establish general robustness.

References

Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction.* Springer, 2 edition, 2009. doi: 10.1007/978-0-387-84858-7. URL https://hastie.su.domains/ElemStatLearn/.